

Histamine, Mast Cells and IgE

Histamine degradation (DAO, HNMT), production, receptors, mast cells and tryptase, IgE and atopy, and skin barrier

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

What was seen. 43 SNPs were read in 26 genes and 7 polygenic scores. 32 SNPs come from the chip, 7 from the imputation ($DR2 \geq 0.8$) and 4 were not found with sufficient quality (the null FLG R501X and 3 pQTLs from the tryptase).

Degradation of histamine slightly reduced, with nothing serious. Sérgio F is heterozygous in the two major functional variants of **DAO (Thr16Met** rs10156191 C/T and **His645Asp** rs1049793 C/G) and in the main variant of **HNMT (Thr105Ile** rs11558538 C/T). It does not have the Ser332Phe nor the 691T promoter of DAO. They are functional 3 heterozygous and 0 homozygous: a common pattern, which only has practical interest if there are symptoms after food rich in histamine.

Mastocytes and tryptase: non-evaluable, with tranquilizing signals. The published proxy of H α T (rs111911698) and alleles that increase tryptase or are associated with mastocytosis (rs7350864, GCSAML, CEBPA) are absent. The number of copies of TPSAB1 is not read in the chip.

IgE and atopy: mixed genetics, low scores. It has common risk alleles in **FCER1A** (T/T), **LPP** (G/G), **STAT6** (2), **IL4** -590T, **IL33**, **TSLP** and **HRNR** (1q21.3), but it does not have the IL13 Arg130Gln (high IgE allele with stronger evidence) and has 1 protective allele in HLA-DQ. Polygenic scores are typical or low: IgE p45,7, allergic disease p21,5, atopic dermatitis p9,8, asthma p3.6 (score A) and p31.0 (score B). The basophils are in p91.3 (informative).

Indirect routes. NAT2 slow acetylator (*5/*5, already known) and normal ALDH2, with zero alcohol.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 43 SNPs: 32 of the chip, 7 imputed and 4 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 2 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizygous).
- **Polygenic scores (PGS):** 7 PGS Catalog scores calculated locally with 23,009 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

Percentages in the reference population. The scores with many variants (asthma A, eosinophils, basophils) are the most reliable; IgE has only 7 variants.

Polygenic IgE scores and allergic cells

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Total IgE in blood	PGS000315 (Circ Genom Precision Med 2020)	7 7 (weighted coverage 100%)	45.7	TYPICAL	typical
Eosinophil count	PGS000165 (Cell 2020)	486 607 (weighted coverage 83%)	57.7	TYPICAL	typical
Counting of basophils	PGS000163 (Cell 2020)	139 185 (weighted coverage 77%)	91.3	MORE HIGH	without judgment

Polygenic scores of allergic disease

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Allergic disease (asthma, rhinitis, eczema)	PGS000810 (Nat Genet 2017)	126 136 (weighted coverage 91%)	21.5	TYPICAL	typical
Atopic dermatitis	PGS003486 (J Invest Dermal 2023)	70 71 (weighted coverage 97%)	9.8	LESS LIKELY	favourable
Asthma A	PGS001849 (Am J Hum Genet 2021)	21,984 22,679 (weighted coverage 97%)	3.6	LESS LIKELY	favourable
Asthma score B	PGS004252 (Ecotoxicol Environ Saf 2023)	197 212 (weighted coverage 89%)	31.0	TYPICAL	typical

4. Variants by theme

For each SNP: genotype (strand +), origin (chip or imputed), effect allele and number of copies, judgment and evidence. In SNPs marked "no judgment" the functional direction is uncertain.

Histamine degradation (DAO and HNMT)

Histamine is inactivated by two enzymes: **DAO** (AOC1 gene), outside the cells and mainly in the intestine (it is the one that treats food histamine), and **HNMT**, inside the cells, in the liver, in the skin and in the brain.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
AOC1 (DAO)	rs10156191	Thr16Met	C/T	chip	T (1 copy/s)	unfavourable (light)	Lower DAO activity in vitro; association with NSAID intolerance and migraine	Moderate
AOC1 (DAO)	rs1049742	Ser332Phe	C/C	chip	T (0 copy/s)	favourable	Lower DAO activity	Moderate
AOC1 (DAO)	rs1049793	His645Asp	C/G	chip	G (1 copy/s)	unfavourable (light)	Less activity and less protein DAO in serum (pQTL)	Moderate-strong
AOC1 (DAO)	rs2052129	−691G>T (promoter)	G/G	chip	T (0 copy/s)	favourable	Lower expression of DAO	Moderate
AOC1 (DAO)	rs1005390	intergenic	T/T	chip	G (0 copy/s)	favourable	Increases plasma 1-methyl-histamine (GWAS of metabolites)	Moderate
AOC1/ KCNH2	rs62492368	locus DAO	G/A	imputation DR2=0.99	A (1 copy/s)	without judgment	Less urinary 1-methyl-histamine; uncertain functional significance	Moderate (association)
HNMT	rs11558538	Thr105Ile	C/T	chip	T (1 copy/s)	unfavourable (light)	Thermolabile enzyme and much less HNMT protein (main pathway inside the cells and in the brain)	Strong (biochemistry); moderate (clinical)
HNMT	rs2071048	−1639 (promoter)	C/C	imputation DR2=0.92	T (0 copy/s)	favourable	Unstudied functional effect	Weak

Production and indirect routes

Production by HDC (histidine decarboxylase) and indirect pathways: ALDH2 eliminates alcohol acetaldehyde, which releases histamine; NAT2 acetylates a small part of histamine and many drugs.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HDC	rs2073440	Glu644Asp	T/G	chip	G (1 copy/s)	unfavourable (light)	Associations with allergic rhinitis and asthma in small studies	Weak
HDC	rs17740607	intronic	G/G	imputation DR2=0.99	A (0 copy/s)	favourable	Association with allergic rhinitis (small studies)	Weak
ALDH2	rs671	Glu504Lys	G/G	chip	A (0 copy/s)	favourable	Acetaldehyde accumulates with alcohol and release histamine; irrelevant with zero alcohol	Strong (with alcohol)
NAT2	rs1801280	*5 (Ile114Thr)	C/C	chip	C (2 copy/s)	unfavourable	Slow acetylation (minor histamine pathway; main of several drugs)	Strong
NAT2	rs1495741	Slow acetylator tag	A/A	chip	A (2 copy/s)	unfavourable	GWAS marker of slow acetylator phenotype	Strong

Histamine receptors

The H1 receptors (allergy, pruritus, wakefulness), H2 (gastric acid) and H4 (immune cells). The available SNPs have weak evidence.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HRH1	rs901865	5'UTR	T/C	chip	T (1 copy/s)	without judgment	No effect demonstrated in histamine response	Weak
HRH2	rs2067474	-1018G>A (promoter)	G/G	chip	A (0 copy/s)	favourable	Associations with asthma and inconsistent gastric cancer	Weak
HRH4	rs11662595	intronic	A/A	chip	G (0 copy/s)	favourable	Associations with atopic dermatitis and asthma (small studies)	Weak
HRH4	rs11665084	intronic	C/C	chip	T (0 copy/s)	favourable	Associations with asthma (small studies)	Weak

Mastocytes and tryptase

Tryptase is the marker of mast cells. Hereditary alpha-tryptasemia (HαT) is a variation in the number of copies of TPSAB1 that the chip does not measure; published proxies and pQTLs were used. The KIT D816V mutation of mastocytosis is somatic and does not appear in the germinative DNA.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TPSG1/TPSAB1	rs111911698	Proxy of hereditary alpha-tryptasemia	A/A	imputation DR2=0.95	G (0 copy/s)	favourable	Proxy published from HαT (a cohort). It does not replace the TPSAB1 copy count	Weakness
TPSAB1/TPSD1	rs7350864	pQTL	A/A	imputation DR2=0.87	G (0 copy/s)	favourable	Increases the protein TPSAB1 in serum	Moderate
TPSAB1/TPSD1	rs9937881	pQTL	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T faux copy/s)	without data	More TPSB2/TPSAB1 protein and more systemic mastocytosis	Moderate
TPSG1	rs138118686	pQTL	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	G faux copy/s)	without data	Increases the protein TPSAB1 in serum	Moderate
TPSAB1	rs1060314	pQTL	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T faux copy/s)	without data	Increases tryptase beta-2; T is the common allele	Moderate
GCSAML	rs56043070	intronic	G/G	imputation DR2=0.93	A (0 copy/s)	favourable	Chronic urticaria and higher tryptase	Moderate
CEBPA	rs4616402	intergenic	G/G	imputation DR2=0.94	A (0 copy/s)	favourable	Mastocytosis with KIT D816V	Moderate

IgE and atopy

Total IgE loci, eosinophils and allergic disease (asthma, rhinitis, eczema): via Th2 (IL4, IL13, STAT6), epithelial allarmines (TSLP, IL33) and high affinity receptor of IgE (FCER1A).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FCER1A	rs2251746	promoter	T/T	chip	T (2 copy/s)	unfavourable	Higher total IgE and higher FcεRI in basophils (coding of inconsistent effect between studies)	Moderate-strong
FCER1A	rs2427837	intronic	G/G	chip	A (0 copy/s)	favourable	Higher IgE	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
STAT6	rs1059513	3'UTR	T/C	chip	T (1 copy/s)	unfavourable (light)	More asthma and eosinophils; associated with IgE	Strong
STAT6	rs3024971	intronic	T/G	chip	T (1 copy/s)	unfavourable (light)	More eosinophils and asthma	Strong
STAT6	rs324011	intron 2	C/C	chip	T (0 copy/s)	favourable	IgE and asthma in some studies	Weak
IL13	rs20541	Arg130Gln	G/G	chip	A (0 copy/s)	favourable	More IgE, asthma and atopic dermatitis (the high IgE allele with stronger evidence)	Strong
IL13	rs1800925	-1112C>T (promoter)	C/C	chip	T (0 copy/s)	favourable	More IL-13 and asthma	Moderate
IL4	rs2243250	-590C>T (promoter)	C/T	chip	T (1 copy/s)	unfavourable (light)	More IL-4 and total IgE (moderate effect)	Moderate
IL4R	rs1801275	Gln576Arg	A/G	chip	G (1 copy/s)	unfavourable (light)	IgE and atopy in some studies; inconsistent	Weak
HLA-DQA2/DQB1	rs2858331	HLA class II region	A/G	chip	G (1 copy/s)	favourable	G allele decreases total IgE	Moderate
HLA-DQB1	rs9273349	HLA class II region	T/C	chip	C (1 copy/s)	unfavourable (light)	More asthma	Moderate
HLA-C	rs3130941	HLA class I region	G/G	chip	C (0 copy/s)	favourable	Increases total IgE	Moderate
LPP	rs9865818	intronic	G/G	chip	G (2 copy/s)	unfavourable	More allergic sensitization	Moderate
IL33	rs992969	intergenic (ERMP1/IL33)	A/G	chip	A (1 copy/s)	unfavourable (light)	More asthma and eosinophils (very replicated)	Strong
TSLP	rs1438673	intergenic (WDR36/TSLP)	C/T	chip	C (1 copy/s)	unfavourable (light)	More allergic disease (asthma, rhinitis, eczema)	Strong
IL1RL1 (ST2)	rs950880	intronic	C/A	chip	A (1 copy/s)	without judgment	Decreases soluble sST2; associated with asthma (clinical direction uncertain)	Moderate
EMSY	rs7927894	11q13.5	C/C	chip	T (0 copy/s)	favourable	More atopic dermatitis and asthma	Moderate-strong

Skin barrier

Filaggrin and the locus of eczema 1q21.3. The nulls of FLG are the major genetic factor of atopic dermatitis.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FLG	rs61816761	R501X (null)	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	The so-called copy/s	without data	Filaggrin main nucleus: atopic dermatitis, dry skin, ichthyosis	Strong
HRNR/FLG	rs12123821	locus 1q21.3	C/T	chip	T (1 copy/s)	unfavourable (light)	Atopic dermatitis (very replicated) and more IgE	Strong

5. Axis reading

Axis	Genetic reading
Degradation (DAO, HNMT)	Slightly reduced: 3 functional heterozygotes, 0 homozygotes
Production (HDC)	Neutral (weak evidence)
Receptors (H1, H2, H4)	No sign
Mastocytes and tryptase	Not assessable in the chip; tranquilizer proxies
IgE and atopy	Common risk alleles, but typical to low polygenic scores
Skin barrier	Normal to slightly vulnerable (1 allele 1q21.3; R501X not read)

What does this mean in practice: genetics suggests a histamine degradation capacity slightly below average, with no strong predisposition to allergy. It only gains relevance if there are symptoms (cephaly, flushing, nasal congestion, itching, diarrhea or palpitations) after foods rich in histamine, alcohol or certain drugs. The skin deserves attention by seborrheic dermatitis and isotretinoin, which weigh more than these genes.

6. Favourable

- **DAO without the worst function variants:** Ser332Phe (rs1049742 C/C) and promoter –691T (rs2052129 G/G) absent; without the allele that rises to 1-methyl-histamine (rs1005390 T/T).
- **HNMT:** –1639 promoter without the effect allele (rs2071048 C/C, imputed).
- **Normal ALDH2** (rs671 G/G): with zero alcohol, acetaldehyde is not a source of histamine release.
- **Receptors without risk signal:** HRH2 rs2067474 G/G; HRH4 rs11662595 A/A and rs11665084 C/C.
- **Mastocytes:** without the proxy of hereditary alpha-tryptasemia (rs111911698 A/A, imputation DR2 0.95), without the allele that rises the tryptase rs7350864 (A/A) and without the alleles of chronic urticaria (GCSAML G/G) and mastocytosis (CEBPA G/G).
- **IgE:** without IL13 Arg130Gln (rs20541 G/G) and without the IL13 –1112T promoter; FCER1A rs2427837, STAT6 rs324011, HLA-C rs3130941 and EMSY rs7927894 without the risk allele 1 protective allele in HLA-DQA2/DQB1 (rs2858331 G).
- **Low scores:** Atopic dermatitis **p9.8**, asthma **p3.6** (score A, the largest) and allergic disease p21.5.

7. Unfavourable

- **DAO:** Heterozygous **Thr16Met** (rs10156191 C/T) and **His645Asp** (rs1049793 C/G; less protein DAO in serum).
- **HNMT: Thr105Ile** heterozygous (rs11558538 C/T; thermolabile enzyme).
- **NAT2 slow acetylator** (*5/*5; rs1801280 C/C, rs1495741 A/A): minor route of histamine, but relevant for drugs that release or inhibit DAO (isonazide, hydralazine).
- **IgE and atopy:** FCER1A rs2251746 T/T (more IgE), LPP rs9865818 G/G (more sensitization), STAT6 rs1059513 and rs3024971 (1 copy each), IL4 -590T (1), IL4R Gln576Arg (1; weak), HLA-DQB1 rs9273349 C (1), IL33 rs992969 A (1), TSLP rs1438673 C (1).
- **Cutaneous barrier** 1 Risk allele in the locus of eczema **HRNR/FLG rs12123821** (T).
- **Production:** HDC Glu644Asp heterozygous (weak).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Chronic Inflammation	DAO and HNMT with 3 alleles of lower function (heterozygous); histamine accumulated maintains low-grade inflammation	unfavourable (light)
Intestinal Barrier dysfunction	Intestinal DAO is the barrier against food histamine (Thr16Met and His645Asp heterozygous)	unfavourable (light)
Intercellular Communication	Th2 and alarmins (STAT6, IL4, IL33, TSLP, FCER1A) with common risk alleles IL13 Arg130Gln absent	mixed
Immunosenescence	Mastocytes without signs of H α T or mastocytosis in proxies; basophils PGS p91.3 (informative)	neutral
Inflamming	Scores of low allergic disease and asthma (p21.5; p3.6)	favourable
Sleep disturbance	Brain histamine (HNMT) remains awake; HNMT Thr105Ile heterozygous	unfavourable (light)
Remodelling of the extracellular matrix	Cutaneous barrier: 1 risk allele in the locus 1q21.3 (HRNR/FLG)	unfavourable (light)
Dysbiosis	Intestinal bacteria produce histamine; with a slightly reduced intestinal DAO, a balanced microbiota and a lot of fiber help	unfavourable (light)

9. Practical suggestions

All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Only if there are symptoms** after food rich in histamine (cured cheeses, sausages, canned fish or not fresh, fermented, tomato, spinach): experiment with 2–4 weeks of poor diet in histamine, with symptoms. Without symptoms, it is not necessary to restrict. *Suggestion (not applied).*
2. **Fish always very fresh or frozen right after fishing** (histamine is formed in the poorly conserved fish). Maintains fish oil in 3.2–4 g/day of EPA+DHA. *Suggestion (not applied).*
3. **Basal serum tryptase** in a routine analysis, once; if it is ≥ 8 ng/mL, request the TPSAB1 copy count (it is the only way to confirm or exclude HxT). *Suggestion (not applied).*
4. **Drugs that inhibit DAO or release histamine** (e.g. isoniazid, which is also acetylated by slow NAT2, some muscle relaxants, opioids such as morphine, iodinated contrasts): warning the doctor of the profile DAO/HNMT + slow NAT2 before using them. *Suggestion (not applied).*
5. **Zero alcohol** (already the rule): alcohol inhibits DAO and releases histamine. *Suggestion (not applied).*
6. **Cofactors of DAO by feeding:** Copper and vitamin C (dry fruits, legumes, cocoa, fruit and vegetables); supplemental copper only with the doctor. *Suggestion (not applied).*
7. **Skin:** Daily hydration and sunscreen (isotretinoin; locus 1q21.3); topical treatment of seborrheic dermatitis according to the plan. *Suggestion (not applied).*
8. **H1 antihistamines of 2.a generation** (cetirizine, bilastin) are the option if there are allergic symptoms; none of these genes changes the choice. *Suggestion (not applied).*

10. Limitations

- It is a consumer chip: SNPs have small effects and nothing here is diagnosed with intolerance to histamine, allergy or mastocytosis.
- Hereditary alpha-tryptasemia is a variation in the number of copies that the chip does not measure; the rs111911698 proxy comes from a single cohort. **KIT D816V** is somatic.
- **Unread with DR2 $\geq$ 0.8:** FLG R501X (rs61816761) and the pQTLs of tryptase rs9937881, rs138118686 and rs1060314 (16p13.3 duplicate region). A previous inference by neighbors gave them as homozygous to the common allele; here they are not read.
- FLG 2282del4 is illegible in the chip; other rare zeros of FLG were not confirmed.
- Conclusions only from genes; do not include blood analysis.

11. Methods and sources

- **Genes and SNPs:** 43 SNPs chosen from the previous analysis of histamine, mast cells and IgE (96 lines), maintaining functional variants and loci with moderate or strong evidence; variants counted by commercial reports without demonstrated functional effect were removed.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report. Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Guidance:** the published effect allele was checked against the local alleles; 2 palindromic SNPs were assumed in the strand +.
- **Polygenic scores:** PGS Catalog, calculated with the imputed genome; percentiles in the reference population.
- **Comparison with previous analysis (02-10-2026):** The 32 genotypes of the chip are equal; the 7 imputed agree with the previous inference.
- **Context Literature:** GWAS Catalog, dbSNP and reviews on DAO/HNMT (Maintz 2006, Comas-Basté 2020), H α T (Lyons 2016) and loci of atopy (Weidinger 2008, Ferreira 2017).

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Histamine degradation (DAO and HNMT)	AOC1 (DAO)	rs10156191	C/T	chip	T	1	strand +	1 copy of risk allele
Histamine degradation (DAO and HNMT)	AOC1 (DAO)	rs1049742	C/C	chip	T	0	strand +	without the risk allele
Histamine degradation (DAO and HNMT)	AOC1 (DAO)	rs1049793	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Histamine degradation (DAO and HNMT)	AOC1 (DAO)	rs2052129	G/G	chip	T	0	strand +	without the risk allele
Histamine degradation (DAO and HNMT)	AOC1 (DAO)	rs1005390	T/T	chip	G	0	strand +	without the risk allele
Histamine degradation (DAO and HNMT)	AOC1/ KCNH2	rs62492368	G/A	imputation DR2=0.99	A	1	strand +	1 copy of effect allele
Histamine degradation (DAO and HNMT)	HNMT	rs11558538	C/T	chip	T	1	strand +	1 copy of risk allele
Histamine degradation (DAO and HNMT)	HNMT	rs2071048	C/C	imputation DR2=0.92	T	0	strand +	without the risk allele
Production and indirect routes	HDC	rs2073440	T/G	chip	G	1	strand +	1 copy of risk allele
Production and indirect routes	HDC	rs17740607	G/G	imputation DR2=0.99	A	0	strand +	without the risk allele
Production and indirect routes	ALDH2	rs671	G/G	chip	A	0	strand +	without the risk allele
Production and indirect routes	NAT2	rs1801280	C/C	chip	C	2	strand +	2 risk allele copies
Production and indirect routes	NAT2	rs1495741	A/A	chip	A	2	strand +	2 risk allele copies
Histamine receptors	HRH1	rs901865	T/C	chip	T	1	strand +	1 copy of effect allele
Histamine receptors	HRH2	rs2067474	G/G	chip	A	0	strand +	without the risk allele
Histamine receptors	HRH4	rs11662595	A/A	chip	G	0	strand +	without the risk allele
Histamine receptors	HRH4	rs11665084	C/C	chip	T	0	strand +	without the risk allele
Mastocytes and tryptase	TPSG1/ TPSAB1	rs111911698	A/A	imputation DR2=0.95	G	0	strand +	without the risk allele
Mastocytes and tryptase	TPSAB1/ TPSD1	rs7350864	A/A	imputation DR2=0.87	G	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Mastocytes and tryptase	TPSAB1/TPSD1	rs9937881	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T	—	—	SNP not found in the data
Mastocytes and tryptase	TPSG1	rs138118686	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	G	—	—	SNP not found in the data
Mastocytes and tryptase	TPSAB1	rs1060314	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T	—	—	SNP not found in the data
Mastocytes and tryptase	GCSAML	rs56043070	G/G	imputation DR2=0.93	A	0	strand +	without the risk allele
Mastocytes and tryptase	CEBPA	rs4616402	G/G	imputation DR2=0.94	A	0	strand +	without the risk allele
IgE and atopy	FCER1A	rs2251746	T/T	chip	T	2	strand +	2 risk allele copies
IgE and atopy	FCER1A	rs2427837	G/G	chip	A	0	strand +	without the risk allele
IgE and atopy	STAT6	rs1059513	T/C	chip	T	1	strand +	1 copy of risk allele
IgE and atopy	STAT6	rs3024971	T/G	chip	T	1	strand +	1 copy of risk allele
IgE and atopy	STAT6	rs324011	C/C	chip	T	0	strand +	without the risk allele
IgE and atopy	IL13	rs20541	G/G	chip	A	0	strand +	without the risk allele
IgE and atopy	IL13	rs1800925	C/C	chip	T	0	strand +	without the risk allele
IgE and atopy	IL4	rs2243250	C/T	chip	T	1	strand +	1 copy of risk allele
IgE and atopy	IL4R	rs1801275	A/G	chip	G	1	strand +	1 copy of risk allele
IgE and atopy	HLA-DQA2/DQB1	rs2858331	A/G	chip	G	1	strand +	1 copy of the protective allele
IgE and atopy	HLA-DQB1	rs9273349	T/C	chip	C	1	strand +	1 copy of risk allele
IgE and atopy	HLA-C	rs3130941	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
IgE and atopy	LPP	rs9865818	G/G	chip	G	2	strand +	2 risk allele copies
IgE and atopy	IL33	rs992969	A/G	chip	A	1	strand +	1 copy of risk allele
IgE and atopy	TSLP	rs1438673	C/T	chip	C	1	strand +	1 copy of risk allele
IgE and atopy	IL1RL1 (ST2)	rs950880	C/A	chip	A	1	strand +	1 copy of effect allele
IgE and atopy	EMSY	rs7927894	C/C	chip	T	0	strand +	without the risk allele
Skin barrier	FLG	rs61816761	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	A	—	—	SNP not found in the data
Skin barrier	HRNR/FLG	rs12123821	C/T	chip	T	1	strand +	1 copy of risk allele